



Surface wettability and pool boiling Critical Heat Flux of Accident Tolerant Fuel cladding-FeCrAl alloys

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ARTICLE INFO

Keywords:

Accident Tolerant Fuel (ATF)

Wettability

Critical Heat Flux (CHF)

DNBR

FeCrAl

ABSTRACT

Surface wettability analysis, including measurements of static (θ_s) advance (θ_a), and receding (θ_r) contact angles, and surface roughness, R_a , are effective parameters used in the literature to predict changes in the pool boiling Critical Heat Flux (CHF). The CHF is an important aspect of the thermal hydraulic performance that needs to be investigated for new Accident Tolerant Fuel (ATF) cladding materials, such as iron-chromium-aluminum alloys (FeCrAl). Surface wettability of FeCrAl samples oxidized under different simulated Light Water Reactor (LWR) water chemistry conditions was measured. These measurements were compared to as-machined and oxidized Zircaloy-4 (Zirc-4), the reference cladding material in LWRs, under the same conditions. Theoretical models were used to predict the pool boiling CHF using surface wettability measurements. Pool boiling experiments were conducted using the same samples to measure the CHF. The measured and predicted CHF data were compared for model validation.

The obtained results showed no significant difference in the measured static contact angle and hence the predicted CHF between the as-machined samples of FeCrAl, 310 SS, and Zirc-4. The contact angles (static, advance, and receding angles) for corroded FeCrAl samples under different LWR water chemistry conditions are lower, and the measured surface roughness values are higher than Zirc-4 corroded under the same conditions. Existing models in the literature predicted higher pool boiling CHF of corroded FeCrAl compared to 310 stainless steel (SS), and Zirc-4. The measured pool boiling CHF for oxidized FeCrAl samples was higher than Zirc-4 samples. The measured and predicted CHF values are in good agreement.

The CHF and the Departure from Nucleate Boiling Ratio (DNBR) were calculated using the Consortium for Advanced Simulation of Light Water Reactors (CASL) subchannel code COBRA-TF (CTF) for a one-eighth model of a 17×17 Pressurized Water Reactor (PWR) fuel assembly. The inlet conditions are consistent with typical PWR values except for the power, which was set 50% higher than is typical to represent possible accident conditions more accurately. The calculated distributions for the CHF and DNBR for oxidized FeCrAl showed significantly higher values throughout the fuel assembly octant compared to as machined Zirc-4 and as machined FeCrAl. These preliminary results show that oxidized FeCrAl may be able to withstand the proposed accident conditions without leading to a boiling crisis.

1. Introduction

In 2011, following the Fukushima Daiichi nuclear power plant accident, the United States Congress instructed the U.S. Department of Energy Office of Nuclear Energy to initiate development of nuclear fuel and cladding materials with potentially enhanced accident tolerance,

known as Accident Tolerant Fuel (ATF) (Zinkle et al., 2014). The main goal of the ATF program is to develop new fuel and cladding materials. These materials would potentially have enhanced accident tolerance to improve the reactor performance and safety characteristics under both normal and severe accident conditions of Light Water Reactors (LWRs.) Among the many desired characteristics of ATF cladding candidate

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<https://doi.org/10.1016/j.nucengdes.2018.08.024>

Received 24 April 2018; Received in revised form 23 August 2018; Accepted 24 August 2018

Available online 28 August 2018

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materials, low oxidation rate, high thermal performance, and stable mechanical properties under normal and accident operational conditions are considered key factors (Koo et al., 2014; Goldner, 2012).

Iron-chromium-aluminum alloys, referred to as FeCrAl, have been under investigation as an ATF candidate material (Field et al., 2017). These investigations have included the evolution of microstructural and mechanical properties under radiation doses relevant to LWRs (Field et al., 2015). These studies show that the chromium content of the FeCrAl alloy significantly impacts the hardening response seen after irradiation. The radiation hardening is dominated by the content of Cr-rich α' precipitates at high chromium content (Edmondson et al., 2016).

Reactor performance and safety characteristics of LWRs with FeCrAl cladding have been studied for both UO₂ fuels and advanced high-density composite fuels (George et al., 2015; Brown et al., 2015). In general, parasitic neutron absorption in iron and chromium harden the neutron energy spectrum, which decreases the achievable discharge burnup and reduces natural resource utilization by approximately 15%. However, full design optimization studies have not been performed for FeCrAl cladding. Studies of severe accident behavior indicate the potential for an increased coping time in a Fukushima-like event, but significant uncertainty in these predictions remain (Robb et al., 2016).

Efforts to study oxidation of different FeCrAl alloys have been reported in the literature (Yamamoto et al., 2015; Pan et al., 2016a,b; Terrani et al., 2014, 2016). The oxidation and mechanical properties of wide range composition, nuclear grade wrought FeCrAl were developed and evaluated. It was found that the tensile properties and the oxidation of the proposed FeCrAl alloys were dependent on the alloy composition. The measured tensile properties were comparable or superior to commercially available FeCrAl. The oxidation under the superheated steam of the proposed FeCrAl alloys was strongly dependent on both Cr and Al contents of the alloy (Yamamoto et al., 2015).

The formation and stability of an oxide layer in FeCrAl fuel cladding material under a high-temperature steam environment was investigated (Pan et al., 2016a,b). The oxidation experiment was conducted at 1000 °C and 1200 °C using FeCrAl alloy having 13% Cr and 5% Al by weight and added elements such as 2.3% Mo, 1% Nb, and trace amounts of C. At 1000 °C, results showed an almost threefold higher oxidation resistance of FeCrAl over Zirc-4 alloy and the formed oxide layer, which consisted of FeO and Al₂O₃, was stable under cyclic cooling. The oxide layer formed at 1200 °C was thicker due to a rapid growth of Al₂O₃ at a higher temperature and unstable under cyclic cooling. The thicker oxide layer experienced cracks and peel off under the cooling process because of the cycling thermal stress.

An out of the pile corrosion study using commercially available FeCrAl alloys under LWR conditions was performed (Terrani et al., 2014, 2016). Samples of FeCrAl composed of 10–18% Cr by weight and 3–5% Al by weight, 310 stainless steel (310 SS), and Zirc-4 were exposed to different water chemistry environments found in LWR primary coolant loops in an autoclave for up to one year. These conditions were hydrogenated water chemistry (HWC), originated or normal water chemistry (NWC) for Boiling Water Reactor (BWR) conditions (7 MPa, and 290 °C), and Pressurized Water Reactor (PWR) water chemistry and operating conditions (15 MPa, and 330 °C). The results concluded inconsequential material loss and reduction in the cladding thickness (~2 mm) after one year of LWR water corrosion in a nonirradiated environment.

The next step for the ATF program is to evaluate the heat transfer characteristics of the newly developed FeCrAl alloys. Pool and flow boiling heat transfer coefficients and Critical Heat Flux (CHF) are key parameters for determining the thermal safety margin for nuclear reactor systems in LWRs (Brown et al., 2016). The CHF is associated with departure from nucleate boiling, a phenomenon in which bubbles coalesce and cover the cooled surface with a vapor blanket. This results in a sharp reduction in the local heat transfer coefficient, and hence a dramatic increase in surface temperature. The induced thermal stress as a result of the rapid temperature increase at CHF conditions would

cause the material fracture. The temperature rise would exceed the maximum allowable temperature at which cladding meltdown would occur (Ali and El-Genk, 2012, 2013; Liang and Mudawar, 2018).

Reported results on the experimentally investigated pool and flow boiling heat transfer and CHF on FeCrAl is limited in the literature. An experimental investigation of the surface characteristics of deposited FeCrAl layers was conducted (Seo et al., 2016). The FeCrAl layers were deposited on top of 316 SS by DC sputtering. Changing the substrate temperature and sputtering time resulted in different surface characteristics of the deposited FeCrAl layer. A significant increase in surface roughness was realized as the substrate temperature increased from 150 to 600 °C and with increased sputtering time up to 6 h. The measured surface roughness ranged from 0.1 to 1.45 μ m and the measured contact angles ranged from 60 to 85° for the smoothest and most rough surfaces, respectively. Pool boiling CHF experiments were conducted using water as the boiling liquid. The CHF was enhanced for the FeCrAl layered surfaces compared to the base 316 SS surfaces. For example, FeCrAl layers deposited at a substrate temperature of 150 °C and a sputtering time of 1 h and FeCrAl deposited at a substrate temperature of 600 °C and a sputtering time of 1 h enhanced CHF by 42% and 32%, respectively. The experimentally measured CHF values were compared to CHF values predicted with existing analytical models using the measured static contact angles on the FeCrAl layered surfaces (Kandlikar, 2001). A partial agreement between the measured and predicted CHF was reported. The difference between measured and predicted CHF was attributed to the use of static contact angle instead of receding contact angle on which the model was developed.

The pool and flow boiling CHF can be measured experimentally, as well as being predicted using some simple surface characteristics measurements (Katto, 1992; Kandlikar, 2001; Chang and Baek, 2003; Pan et al., 2016a,b). Surface measurements include surface roughness and wettability (static and dynamic contact angles). These surface characteristics have shown to greatly impact both pool and flow boiling heat transfer coefficient and CHF (Jo et al., 2011; Hsu and Chen, 2012; O'Hanley et al., 2013; El-Genk et al., 2013; Kim et al., 2016; Jo et al., 2017; Hsu et al., 2017). The impact of contact angle on CHF has been investigated theoretically and experimentally and models were developed and reported in the literature (Zuber, 1959; Kandlikar, 2001; Hsu and Chen, 2012; Liao et al., 2008). These models were compared to experimental CHF data on surfaces of different degrees of wettability.

The impact of surface wettability on nucleate boiling heat transfer and CHF using nano-silica particle coatings was studied by Hsu and Chen (2012). The particle diameter and coating thickness were used to control the surface wettability (static contact angle) of the boiling copper surface between 10° (hydrophilic) to 145° (hydrophobic). Results showed that increasing the surface wettability (lower contact angle) significantly increased the measured CHF. Their measured CHF was not consistent with the predicted CHF using existing models in the literature that account for the effect of contact angle.

Jo et al. (2017), conducted pool boiling experiments on smooth surfaces with nanoscale roughness. The degree of roughness was varied to control surface wettability. Results showed that hydrophobic surfaces (contact angle > 90°) offer higher nucleate boiling heat transfer coefficients compared to hydrophilic surfaces. The effect of surface wettability on nucleate boiling of water over treated copper tubes (5 mm and 7 mm diameters) was investigated (Hsu et al., 2017). The reported results showed that surfaces with lower contact angles (22.5°) offered higher heat transfer coefficients. O'Hanley et al. (2013), investigated the separate effects of contact angle, porosity, and roughness on CHF using pool boiling experiments on a variety of engineered surfaces. They found that porous hydrophilic surfaces enhance CHF more than hydrophobic surfaces, and surface roughness has minimal impact on CHF.

Roughened copper surfaces have shown higher nucleate boiling heat transfer coefficients of dielectric liquid by El-Genk et al. (2013). The surface roughness was varied between 0.039 μ m (smooth) to

1.79 μm (rough). The reported results showed that increasing surface roughness decreased the static contact angle from 78° for the smooth surface (0.039 μm) to 48° on the roughened surface (1.79 μm). In addition, they reported 50% higher heat transfer coefficient and 40% higher CHF for roughened surfaces versus smooth surfaces. Increasing surface roughness has been shown to increase the wetted surface area which increased the CHF (El-Genk et al., 2013). Kim et al. (2016) investigated the effects of surface roughness on copper surfaces having moderate wettability on the heat transfer coefficient and CHF. The roughness varied from 0.041 to 2.36 μm using sanding papers of different grit. The corresponding static contact angle decreased from 62° to 55° as the surface roughness increased. Results showed that increasing surface roughness (decreasing contact angle) increased both the heat transfer coefficient and CHF by 100%.

In summary, results showed that exposure of FeCrAl cladding to different water chemistry environments and operating conditions, such as high pressure and temperature simulating LWR conditions, resulted in a remarkable change in its surface characteristics, including surface roughness and wettability. These surface characteristics have shown to remarkably affect the heat transfer performance of both pool and flow boiling CHF data. Thus, evaluating pool and flow boiling CHF on FeCrAl surfaces oxidized under LWR conditions would provide more reliable information for the reactor safety community. Conducting experiments to evaluate CHF under actual LWR conditions is quite a challenge on a university lab scale because of limited capabilities and safety concerns. Using existing models for predicting pool boiling CHF data could provide a good insight of CHF data for FeCrAl alloys.

The main objective of this research is to present the pool boiling CHF for FeCrAl alloys with different surface characteristics and wettability. The measured CHF values are compared with several predictions from theoretical models in the literature (more details on these models are discussed in the results section) that account for surface wettability. The surface wettability measurements including surface roughness, static contact angles, and receding contact angles are collected from samples exposed to different operating and chemical conditions in an autoclave for an extended period (one year) to account for the surface characteristic changes due to material oxidation under simulated LWR reactor conditions. These samples are the same used in a corrosion study conducted by Terrani et al. (2016). Surface wettability is not the sole factor that could impact the pool boiling CHF but other properties such as thermophysical properties are also important, which is out of this work scope. Thus, these early results are limited in scope, but it provides insights to be integrated with other progressing research efforts for informing or establishing safety limits.

2. Measurement samples

The surface roughness and wettability measurements were conducted for samples prepared and used in an earlier corrosion study. These samples were autoclaved at General Electric Global Research lab. This study was designed to explore the effect of major alloying elements (i.e., Cr and Al) on corrosion behavior of different alloys under simulated LWR environmental conditions. Coupons of an austenitic 310 SS and distinct compositions of FeCrAl samples were exposed alongside samples of Zirc-4 as a reference to immersion testing for one year.

Table 1
Elemental analysis (W%) of selected samples investigated in the immersion test (Terrani et al., 2016).

Alloy ID	Fe	Cr	Zr	Ni	Al	Other elements
Zirc-4	0.2	0.1	98.2	–	–	Si
310 SS	51.9	25.4	–	19.5	–	Mo, Si
Fe12Cr5Al	83.6	12	–	–	4.4	Y, Si
Fe13Cr4Al	83	13	–	–	3.9	Y

Table 2

Autoclave conditions for immersion tests (Terrani et al., 2016).

Autoclave	Chemistry	Temp. (°C)	pH	Pressure (MPa)
PWR	3.57 ppm H ₂	330	7.2	15
BWR-HWC	0.3 ppm H ₂	290	5.6	7
BWR-NWC	1.0 ppm O ₂	290	5.6	7

Table 1 lists the major element composition of testing samples used in the immersion tests (Terrani et al., 2016).

The immersion tests were conducted in a 1-gallon (3.78 L), 316 SS autoclave using purified water that was initially equilibrated with a mixture of Ar, O₂, and H₂ to establish the desired water chemistry. The details of the environmental conditions during the simulated PWR, BWR-HWC, and BWR-NWC conditions are provided in Table 2. More details on experimental setup and procedure, and results of immersion tests can be found in the literature (Terrani et al., 2016).

The wettability measurements (roughness, static, advance, and receding contact angles) were collected for three samples of each alloy listed in Table 1 and each sample was oxidized under one of the three different LWR conditions listed in Table 2 (total of 12 coupons). In addition, the surface roughness and contact angles were also measured for as machined 310 SS, Zirc-4, and FeCrAl samples to explore the difference in the measured surface wettability between oxidized and bare (fresh with no oxidation) samples.

3. Measurements setup and procedure

3.1. Instrumentation description

The surface roughness and contact angles are measured with a surface profilometer and contact angle Goniometer, respectively. The surface profilometer (Dektak 150, Fig. 1a) is a stylus type with 1 mg force to measure surface peaks and valleys. The average roughness, Ra, of each scan is calculated from the surface topology measurements using the Dektak software. The machine can measure samples up to 5 in. in diameter. The system has a 1-millimeter standard vertical range with up to 120,000 data points per scan. The machine is capable of measuring surface flatness, waviness, and radius of curvature as well as characterizing thin-film stress on wafers.

The contact angle Goniometer is a device that can measure contact angle, surface tension, and surface energy. The facility is fitted with an automated dispensing system to control the droplet volume and ensure measurement repeatability. The automated tilting base is a fully controlled base to change the specimen tilting angle and measure both advance and receding contact angles, θ_A and θ_r , respectively. The machine is also equipped with a temperature-controlled hot chamber for conducting measurements at an elevated temperature up to 300 °C (Fig. 1b).

3.2. Measurement procedure

The American Society of Testing Materials (ASTM) provides a fully detailed standard procedure for measuring contact angle (ASTM, 2008, 2013). The procedure summarizes the necessary steps for obtaining consistent results on contact angle measurement using a liquid droplet on the substrate. The number of droplets, droplet volume, and measuring time are specified in the ASTM standards. The contact angle can be quantified by averaging the measured angle of three different droplets on the same surface and each has a volume less than 20 μL . According to the standard, the measuring time from the moment of attaching the droplet to the surface should not exceed 30 s to avoid droplet volume change by evaporation or diffusion into the substrate.

The visual inspection has shown differences in the color grade between the two faces of the same oxidized sample, and between different

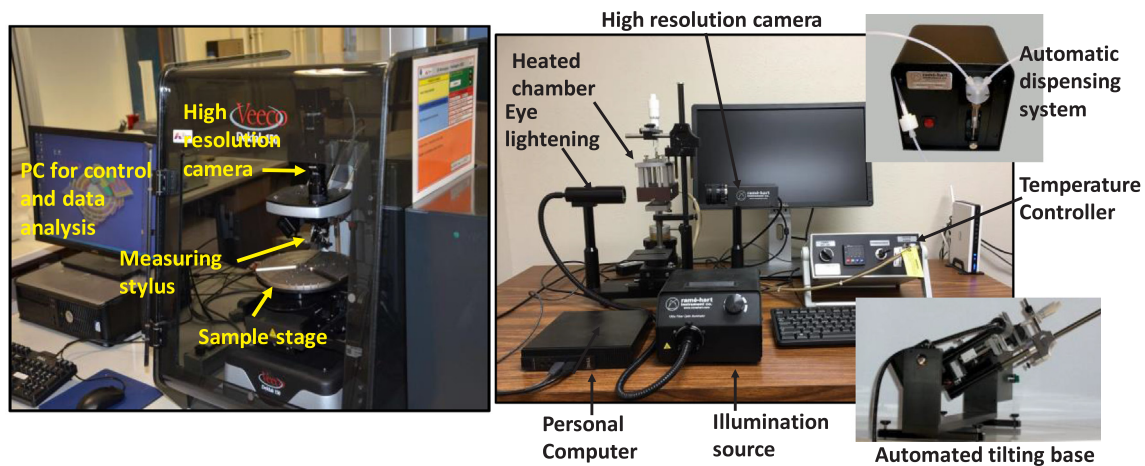


Fig. 1. Measurements facilities (left) Dektak 150 surface profilometer (right) contact angle Goniometer.

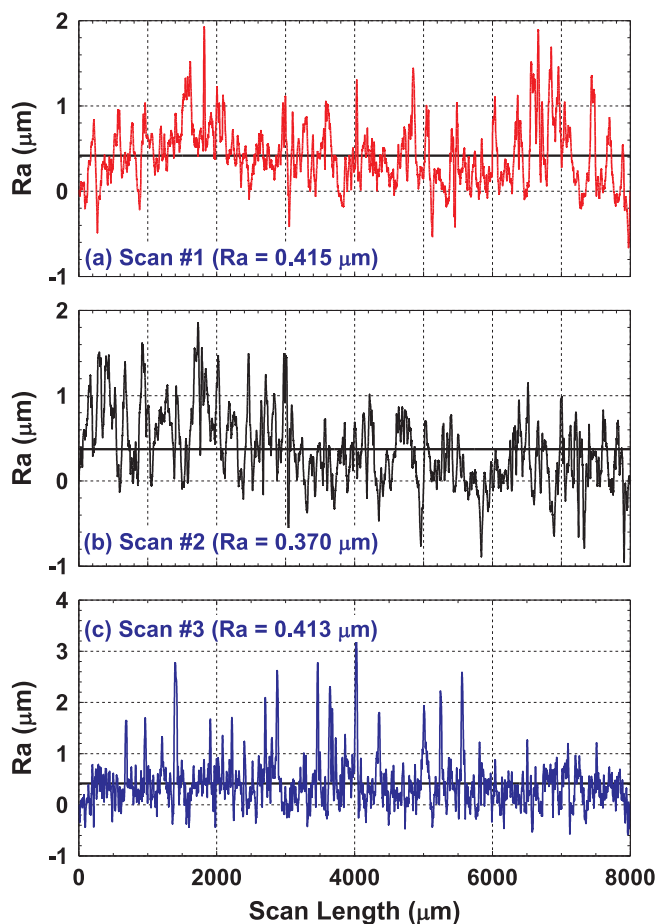


Fig. 2. Individual stylus scan data on Zirc-4-PWR sample (a–c).

samples used in the current study. This color grade indicates the non-uniformity of the oxide layers formed during the corrosion study in the autoclave. Thus, averaging three droplets is not deemed sufficient to present the most accurate contact angles of these samples. To increase the confidence in the representative contact angle of each sample, the measurements were conducted following the ASTM procedure, but with a larger number of droplets. The number of droplets was enough to cover both faces of an individual sample, which varied per sample size between 5 and 8 droplets per sample face. For each sample, the individual droplets' contact angles are presented for each face as well as an average value for all droplets' contact angles on both faces.

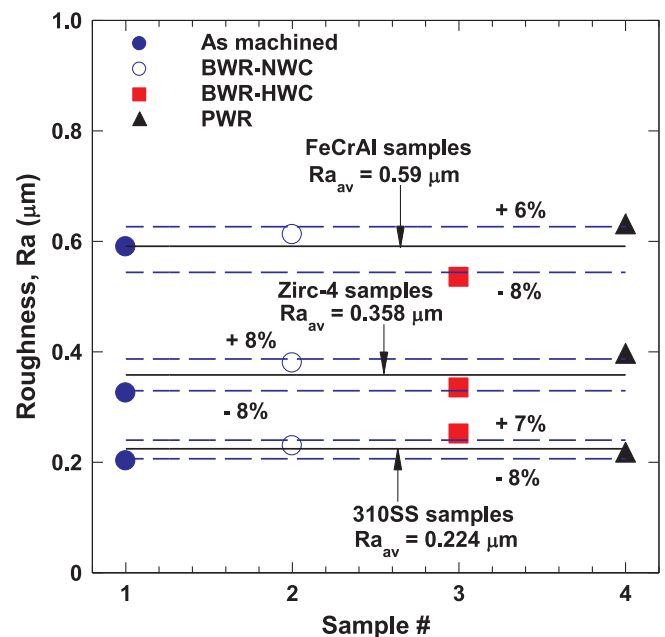


Fig. 3. Measured average surface roughness, R_a , for all samples.

Deionized water was used for this series of measurements with a droplet volume ranging between 16 and 20 μL and the contact angle was recorded 60 times within 30 s. For advance and receding contact angle, the same procedure is valid and followed except that the automated tilting base is rotated from 0 to 90° and both advance and receding angles are recorded (ASTM, 2008, 2013).

For the surface roughness measurement, the Dektak 150 stylus profiler was used to measure the peaks and valleys of the surface topology and calculate the surface roughness of different samples. Three standard scans were performed for each sample. The scan length is 8 mm with 1.33 $\mu\text{m/s}$ scan rate. The collected data from each scan is analyzed using the Dektak software to determine the average roughness, R_a , of each scan. The three calculated roughness values of each sample are averaged and presented in the surface roughness results section.

4. Measurements results

4.1. Surface roughness

As described in the surface roughness procedure, each sample has

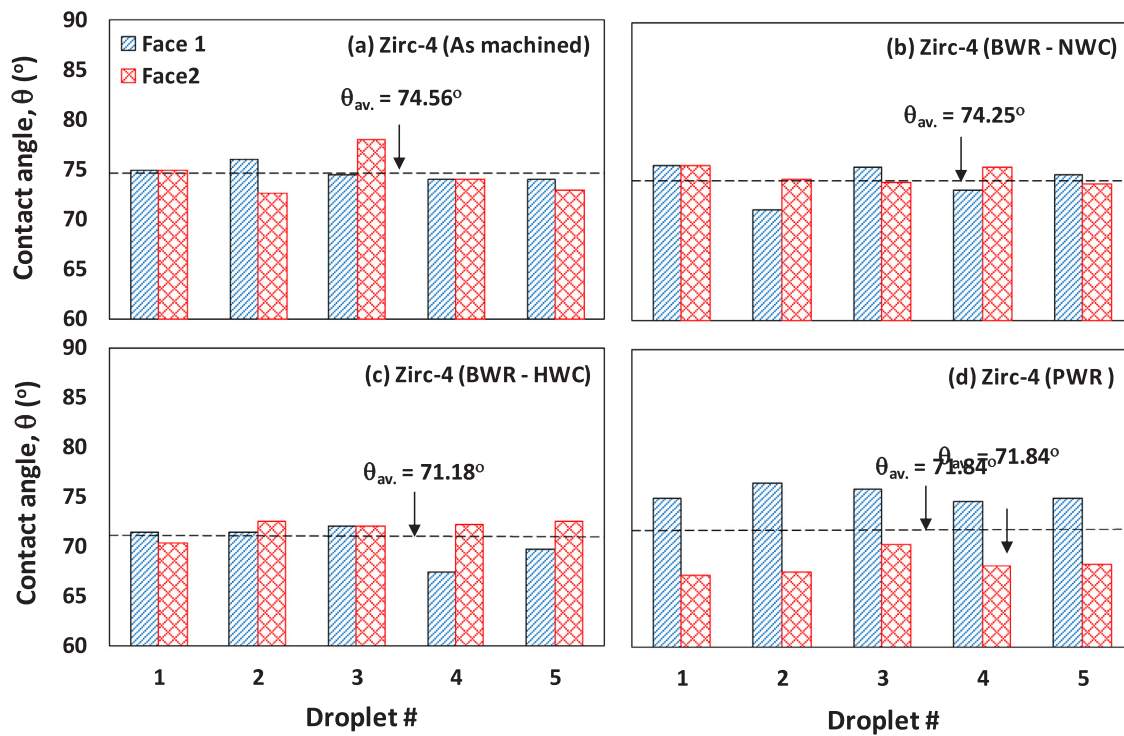


Fig. 4. Measurements of static contact angle on both faces of Zirc-4 samples (Ali et al., 2017).

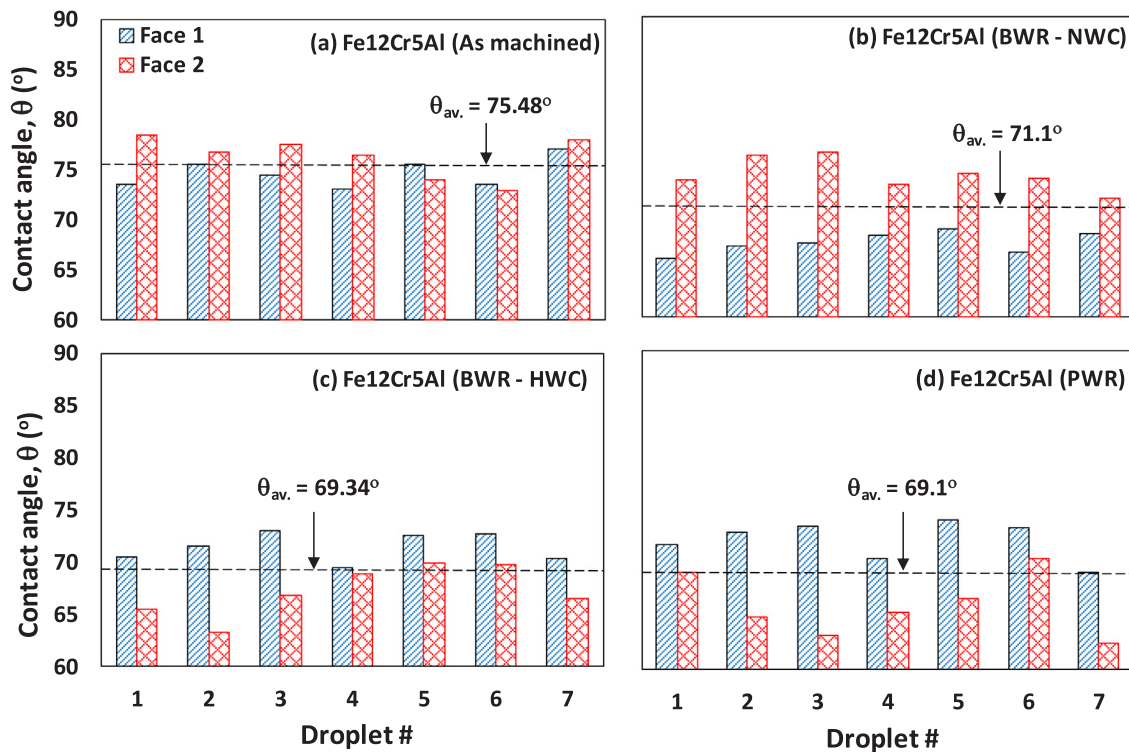


Fig. 5. Measurements of static contact angle on both faces of Fe12Cr5Al samples (Ali et al., 2017).

three different scans with an 8-mm length. Scan routes were selected to cover most of the sample's surface area and avoid any surface scratches. As an example, Fig. 2a–c present the three different scans on a single Zirc-4 sample oxidized under PWR conditions. In each figure, the average value, R_a , presented is the arithmetic mean of the roughness measurements along each scan calculated by the Dektak following the equation, $R_a = \frac{1}{n} \sum_{i=1}^n |y_i|$. The average value of the calculated R_a

values (0.415, 0.37, and 0.413 μm) is $\sim 0.4 \mu\text{m}$. The deviation between the average and the individual values is $< 7\%$ for all specimens. Thus, using arithmetic average of individual measurements is a representative value for each specimen R_a as shown in Fig. 3.

The measured surface roughness of FeCrAl, Zirc-4, and 310SS samples experienced a slight difference between the as machined and oxidized samples regardless of the oxidation conditions (BWR-NWC,

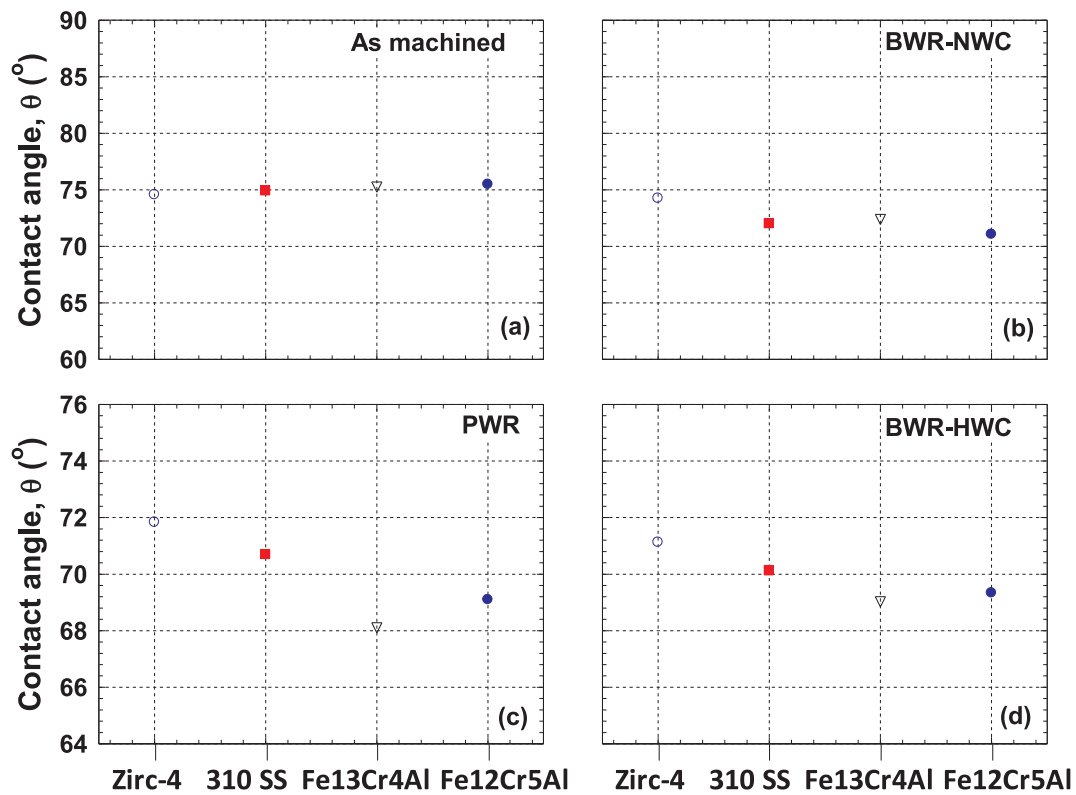


Fig. 6. Comparison of static contact angle for all samples.

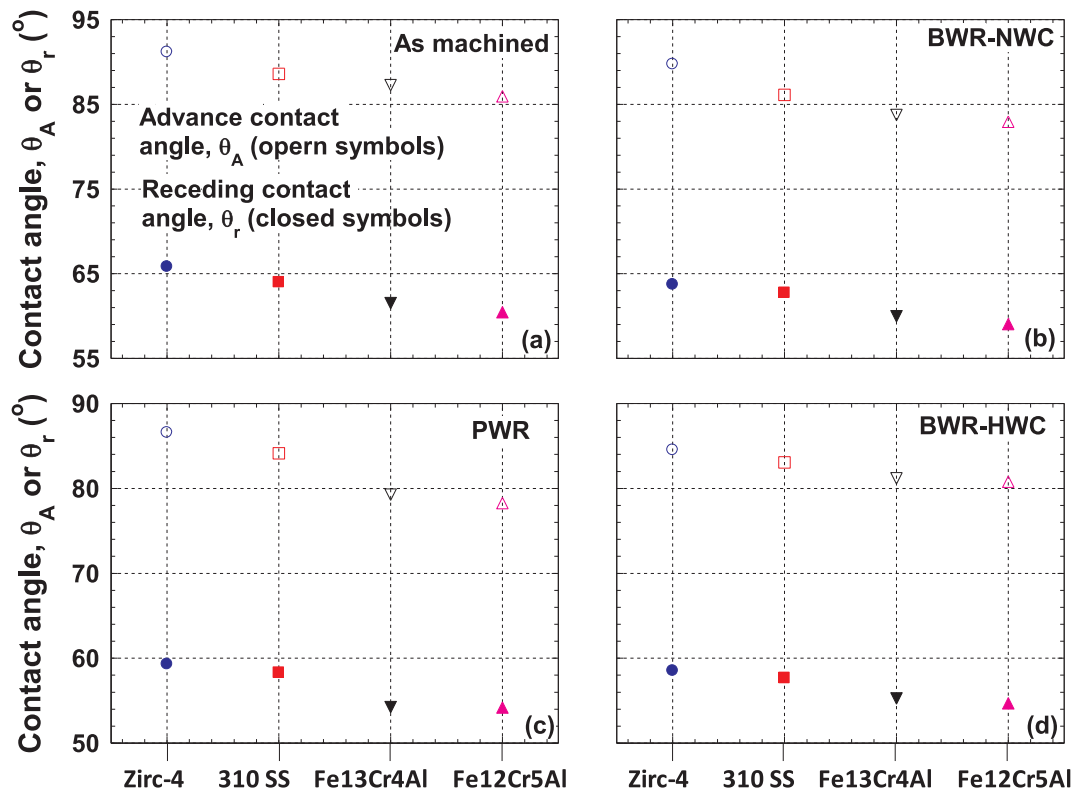


Fig. 7. Comparison of advance and receding contact angles for all samples.

BWR-HWC, and PWR) as shown in Fig. 3. Almost no difference in surface roughness was recognized between Fe13Cr4Al and Fe12Cr5Al samples. The Fe13Cr4Al samples have roughness values of 0.59, 0.61, 0.54, and 0.63 μm for as machined, BWR-NWC, BWR-HWC, and PWR

oxidized samples, respectively. These values are within $\pm 8\%$ or less of an average roughness value of 0.59 μm . The average surface roughness for Zirc-4 and 310 SS samples are 0.358 and 0.224 μm , respectively (Fig. 3). These results show that FeCrAl alloys have a higher roughness

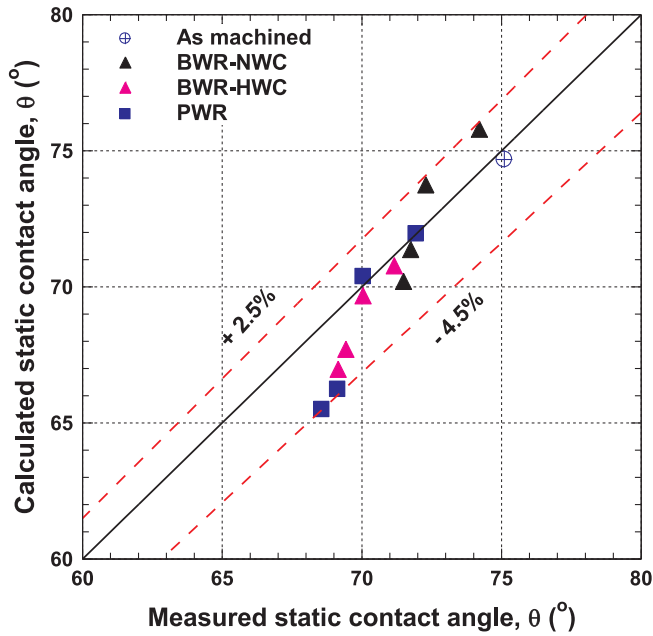


Fig. 8. Comparison between measured and back-calculated static contact angle.

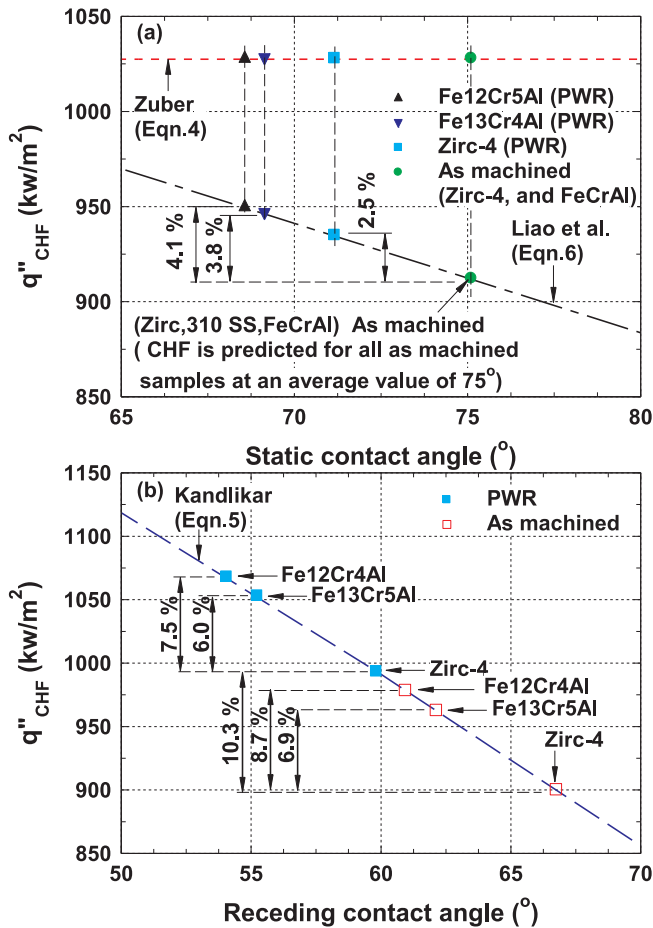


Fig. 9. Predicted CHF based on contact angle measurements (a) Eqs. (4) and (6) (b) Eq. (5).

value compared to Zirc-4 and 310 SS regardless of the autoclave simulated LWR testing conditions.

4.2. Static, advance, and receding contact angles

Samples of the static contact angles measurements of different Zirc-4 and Fe12Cr5Al coupons are presented in Figs. 4 and 5. The selected number of measurements was enough to cover the whole surface area of each sample. For each sample, the measured droplets contact angles on each face are presented by different hatched bars as shown in these figures. The average value of the measured contact angle on both faces of the individual sample is presented by the dashed line in the same figure, which is used to compare the contact angles between different samples. For example, Fig. 4a shows the measured contact angles of 5 different droplets on each face of the as-machined Zirc-4 sample. Two assorted styles of hatched bars present these 10 measurements, with each presenting one face of the sample. The dashed line at 74.56° presents the average of the 10 samples. Fig. 4b–d each show 5 different droplet contact angle measurements and calculated average contact angle values for Zirc-4 samples tested under BWR-NWC, BWR-HWC, and PWR, respectively. Fig. 5 presents the measured contact angle data of Fe12Cr5Al samples.

As shown in Figs. 4 and 5, the difference between the measured contact angles on face 1 and 2 is the minimum for as-machined samples. This difference is more noticeable in the oxidized samples due to the non-uniform surface oxidation. The same observation was recorded for 310 SS and Fe13Cr4Al, as machined and oxidized samples (not presented). The average contact angle values are fitted, and the fitting curves are compared in Fig. 6. For the as machined samples and samples oxidized in the autoclave under simulated LWR conditions. Following the same procedure, the advance and receding contact angles are measured for all samples. The measured advance and receding angle values are compared in Fig. 7.

An expression for the line energy model associated with the solid, liquid, and gas interface line was developed by Tadmor (2004). The model consists of three equations (Eqs. (1)–(3)) that related the drop volume, the interfacial energies, and the advance and receding contact angles to the equilibrium line energy. This expression can back-calculate the equilibrium static contact angle, θ , as a function of the advance, θ_A , and receding, θ_r , contact angles (Tadmor, 2004).

$$\theta = \cos^{-1} \left(\frac{\Gamma_A \cos \theta_A + \Gamma_r \cos \theta_r}{\Gamma_A + \Gamma_r} \right) \quad (1)$$

$$\Gamma_A = \left(\frac{\sin^3 \theta_A}{2 - 3 \cos \theta_A + \cos^3 \theta_A} \right)^{1/3} \quad (2)$$

$$\Gamma_r = \left(\frac{\sin^3 \theta_r}{2 - 3 \cos \theta_r + \cos^3 \theta_r} \right)^{1/3} \quad (3)$$

To ensure the accuracy of the measured advance and receding contact angles, the measured values (Fig. 6) were implemented in the line energy model (Eqs. (1)–(3)) and the corresponding static contact angles were calculated. Fig. 8 compares the measured static contact angles and the back-calculated values using the line energy model. The results show a good agreement between the measured and calculated values.

4.3. Pool boiling CHF prediction

Theoretical models to predict pool boiling CHF were reported in the literature (Zuber, 1959; Kandlikar, 2001; Liao et al., 2008). Zuber (1959), developed an equation to predict the CHF from the hydrodynamic instability model without accounting for surface characteristics. The model was a function of the thermo-physical properties of the boiling liquid.

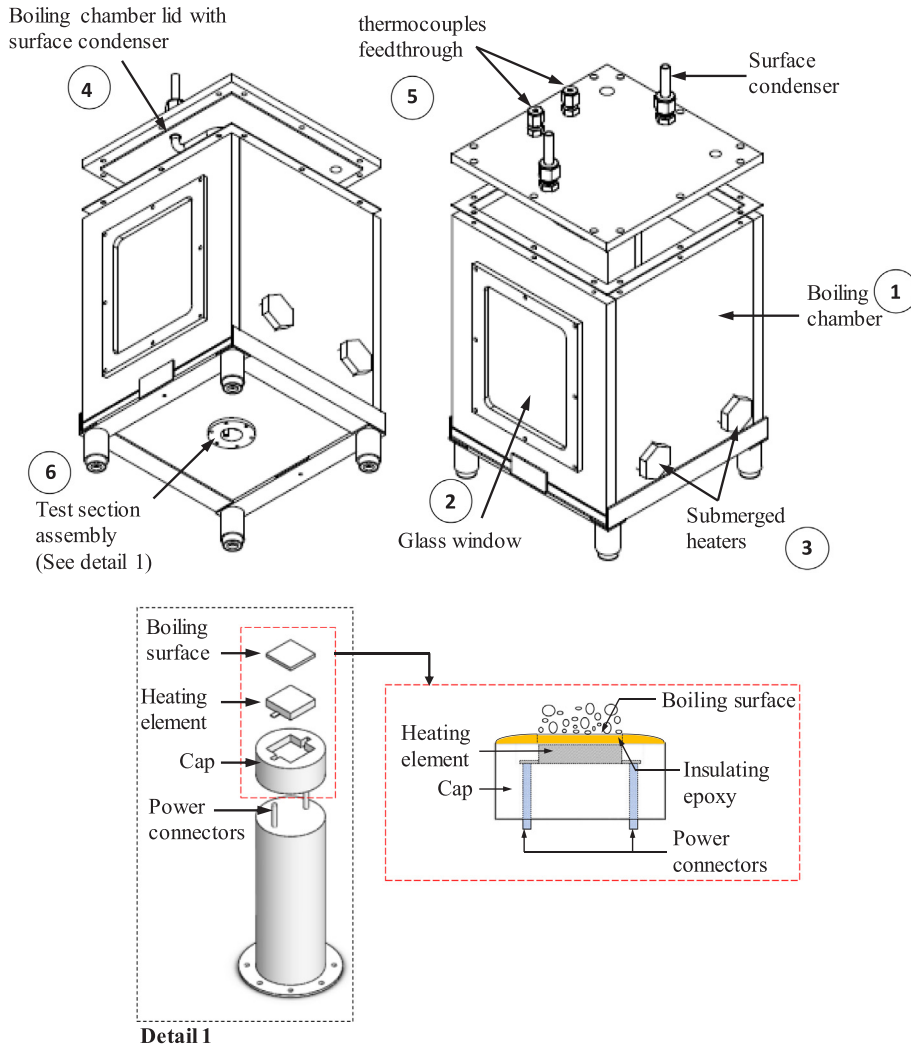


Fig. 10. Schematic of the pool boiling facility.

$$q_{CHF} = Ch_{lg}\rho_g^{0.5} [\sigma g(\rho_l - \rho_g)]^{0.25} \quad (4)$$

Kandlikar (2001), improved Zuber's correlation to predict the CHF in saturated pool boiling considering the impact of surface wettability. A model accounting for receding contact angle, surface orientation, and sub-cooling effects was developed. The model was validated by comparing the predicted CHF values with a wide range of experimental data from the literature. The model was in agreement with experimental data within an error of 21.59%:

$$q_{CHF} = h_{lg}\rho_g^{0.5} [\sigma g(\rho_l - \rho_g)]^{0.25} \left(\frac{1 + \cos\theta_r}{16} \right) \left[\frac{2}{\pi} + \frac{4}{\pi} (1 + \cos\theta_r) \cos\phi \right]^{0.5} \quad (5)$$

Liao et al. (2008), fitted a correlation for CHF taking the effects of both orientation and static contact angle into account (the equation is presented for upward facing orientation, $\phi = 0^\circ$):

$$q_{CHF} = h_{lg}\rho_g^{0.5} \left[1 + \frac{55 - \theta}{100} (0.56) \right] [\sigma g(\rho_l - \rho_g)]^{0.25} \quad (6)$$

In Eqs. (4)–(6), C is a constant, h_{lg} is the liquid latent heat, ρ_l and ρ_g are the liquid and gas density, σ is the liquid surface tension, θ_r and θ are the receding and static contact angles respectively, and ϕ is the surface inclination angle. These models are used to predict the CHF of different samples of Zirc-4, and the ATF candidate, FeCrAl. Fig. 9a shows the predicted CHF using Zuber (1959) and Liao et al. (2008) models and Fig. 9b for the predicted CHF using Kandlikar (2001)

model.

The predicted CHF using Zuber (1959) is about 1027 kW/m² for all samples (the model is independent of the contact angle). Based on the current static contact angle measurements (Fig. 6), Liao et al. (2008) model would predict higher pool boiling CHF for lower contact angle (higher wettability). The oxidized FeCrAl cladding has a lower measured contact angle compared to the oxidized Zirc-4 and/or the as machined samples. The as machined samples (Zirc-4, 310 SS, and Fe-CrAl) have the same static contact angles (Fig. 6) and hence equal predicted CHF using Liao et al. (2008) model (Fig. 9a). The predicted CHF for oxidized Fe12Cr5Al, Fe13Cr4Al, and Zirc-4 under PWR conditions are 950, 946.7, and 935 W/m², respectively. These values are 4.1%, 3.8%, and 2.5% higher compared to predicted CHF for as machined samples (Fig. 9a).

Based on the measured receding and advanced contact angles presented in Fig. 7, Kandlikar (2001), (the highest slope model) predicted higher CHF compared to Liao et al. (2008). The predicted CHF for the as-machined Fe12Cr5Al and Fe13Cr4Al are 6.9% and 8.7% higher compared to predicted CHF for as-machined Zirc-4. These values are increased to 6.0% and 7.5% for oxidized Fe12Cr5Al, and Fe13Cr4Al compared to oxidized Zirc-4 under the same simulated reactor conditions (PWR) as shown in Fig. 9b.

To validate the higher predicted CHF for as machined and oxidized FeCrAl compared to Zirc-4, pool boiling experiments are conducted for selected samples to measure the CHF.

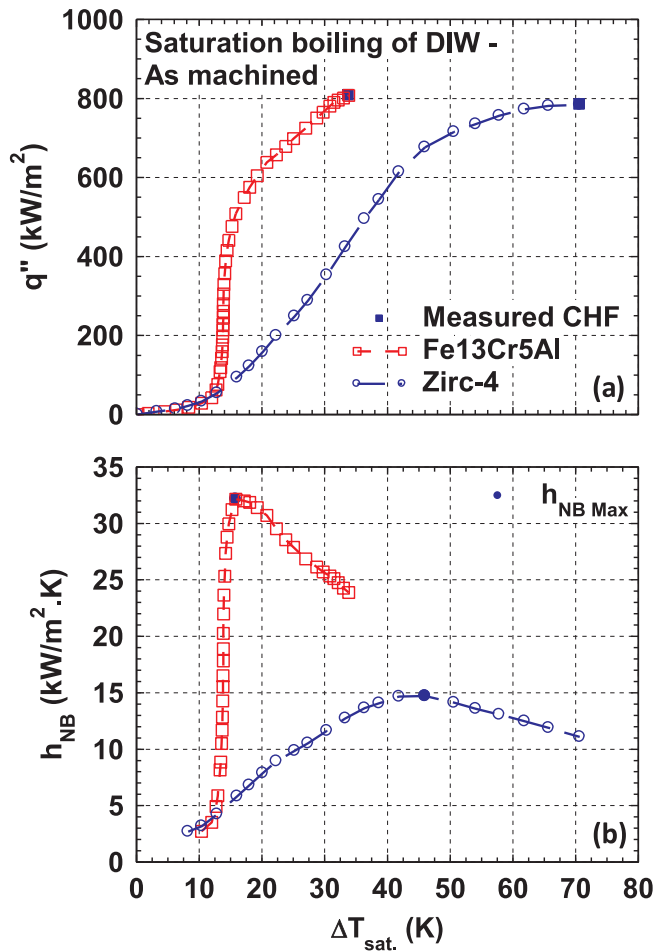


Fig. 11. Pool boiling and heat transfer coefficient curves for as-machined samples.

4.4. Pool boiling CHF measurements

4.4.1. Experimental facility

The pool boiling experimental facility is shown in Fig. 10. The facility is a $180 \times 150 \times 230$ mm³ stainless steel boiling chamber (1). The chamber has front and rear windows for process visualization (2). The chamber is fitted with two 400-Watt submerged heaters for rapid heating and degassing (3). Two 200-Watt surface heaters are attached to the chamber sides (not shown) to keep constant water temperature during the experiments. The chamber lid is fitted with a surface condenser (4) to condense the generated vapor and keep the water level and chamber pressure constant during the experiment. The chamber lid has two feed-through holes (5) for installing thermocouples to measure water and vapor space temperatures. The test sectional assembly (6) is made of poly-ether-ether-ketone (PEEK) and is installed through the bottom of the chamber.

The boiling surface is attached to a heating element using high thermal conductive epoxy and both are inserted into the assembly cap. The boiling surface attached to the cap is externally insulated with two-component epoxy to ensure that all heat supplied by the underneath heating element is dissipated and removed by nucleate boiling from only the top face of the boiling surface but not its sides. The cap has two small feed-through holes to allow for the power connectors going through the PEEK column to get in contact with the heating element. The power connectors through the PEEK column are connected to a DC power supply from outside the boiling chamber. Voltage, current, and temperature signals were collected with a data acquisition system (DAQ) from Measurement Computing. The boiling surface temperature

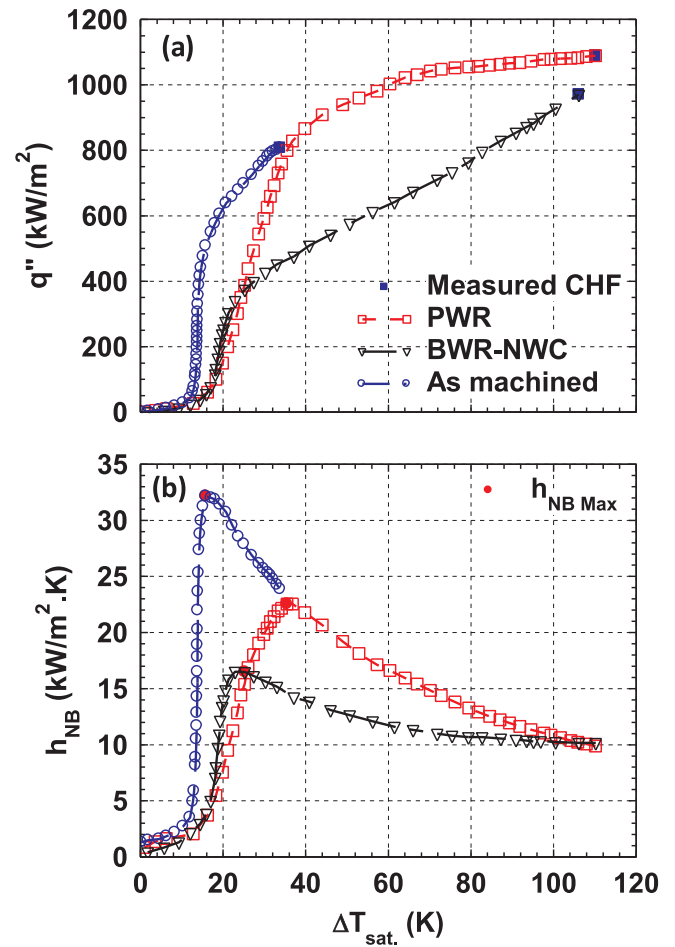


Fig. 12. Pool boiling and heat transfer coefficient curves of different Fe13Cr4Al samples.

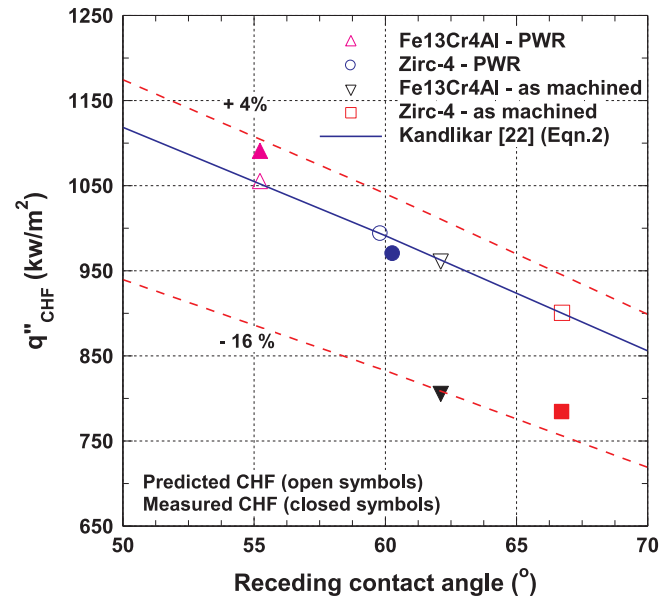


Fig. 13. Comparison between measured and predicted CHF.

was measured with a K-type thermocouple (TC). The bulk temperature of the water pool was also measured with a K-type thermocouple (TC) having a 1-mm sheath diameter inserted through the chamber lid. The nominal uncertainty of the TCs was ± 1.1 °C according to the

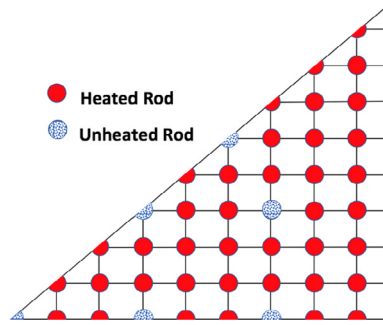


Fig. 14. PWR fuel assembly octane.

Table 3

Fuel rod input parameters and key results.

Assembly	Zirc-4 (as machined)	FeCrAl (as machined)	FeCrAl PWR
Fuel Pellet Diameter (mm)	8.192	8.736	8.736
Gap Thickness (mm)	0.082	0.082	0.082
Cladding Outer Diameter (mm)	9.500	9.500	9.500
Cladding Thickness (mm)	0.572	0.300	0.300
Maximum Heat Flux (kW/m ²)	1442.6	1442.6	1442.6
Minimum Critical Heat Flux (kW/m ²)	245.6	247.6	329.4
Minimum DNBR	0.90	0.91	1.16
Maximum Cladding Outer Surface Temperature (°C)	364.4	350.6	347.2

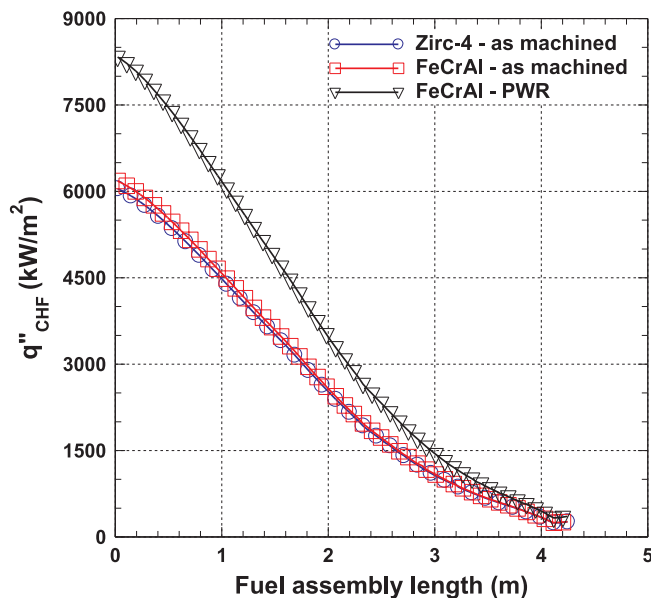


Fig. 15. CHF axial variation for Zirc-4 (as machined), and Fe13Cr5Al (as machined and PWR).

manufacturer, OMEGA Engineering.

4.4.2. Pool boiling CHF results

Samples of Zirc-4 and Fe13Cr4Al are selected for the pool boiling experiment and CHF measurements. These samples are the same samples used in the surface roughness and surface wettability measurements. A total of four samples, each 10x10 mm in size, including two as machined and two Zirc-4 and Fe13Cr4Al samples oxidized under simulated PWR conditions are tested. The pool boiling and heat transfer coefficient curves of different Zirc-4 and Fe13Cr4Al samples are presented in Fig. 11. The results in Fig. 11a show that the CHF for as

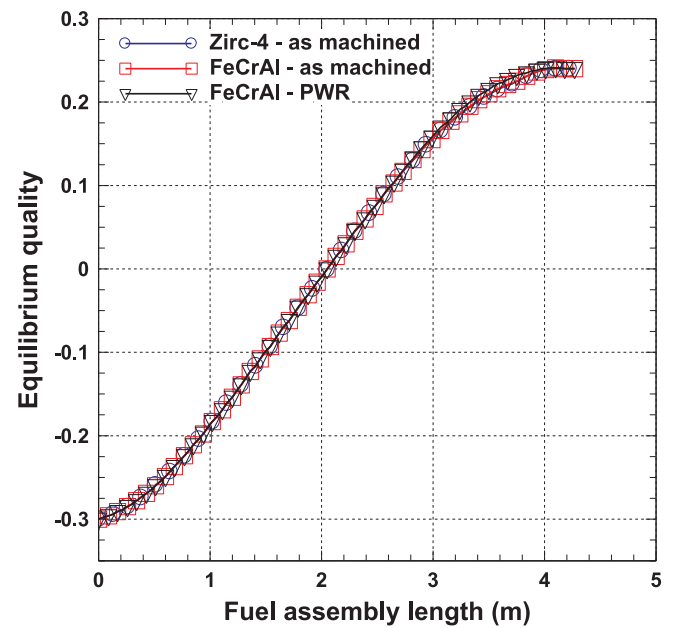


Fig. 16. Axial variation of equilibrium quality.

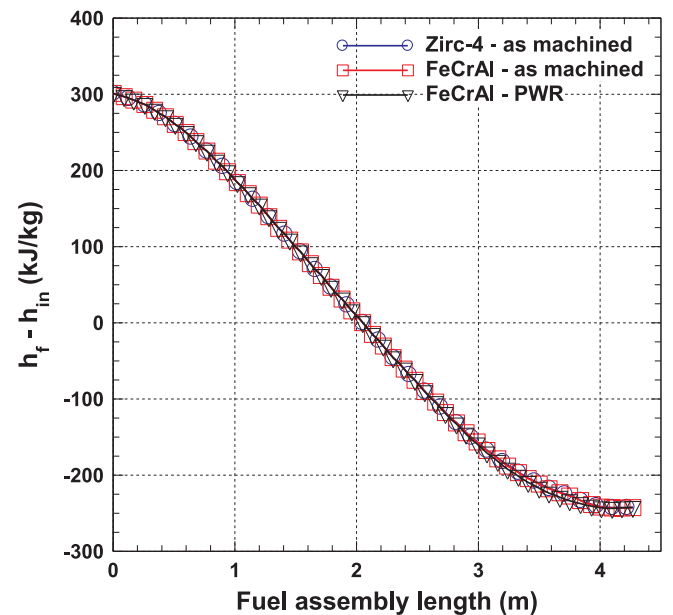


Fig. 17. Axial variation of the difference between saturation enthalpy and actual enthalpy.

machined Fe13Cr4Al (807.5 kW/m²) is almost as the measured CHF of as machined Zirc-4 (784.1 kW/m²). The heat transfer coefficient on Fe13Cr4Al is much higher as indicated by the steep pool boiling curve compared to Zirc-4 and shown in Fig. 11b.

Fig. 12a and b compare the pool boiling and heat transfer coefficient curves for three different Fe13Cr4Al samples: as machined and oxidized samples under simulated PWR and BWR-NWC conditions. As shown in the figure, the presence of the oxide layer increases the measured CHF compared to as machined (no oxidation) samples, but CHF occurs at much higher surface superheat ($\Delta T_{sat.}$). The measured CHF values are 1088.7, 970.2, and 807.5 kW/m² for Fe13Cr4Al samples oxidized under PWR conditions, sample oxidized under BWR-NWC conditions, and as machined sample, respectively. The figure also shows that the as machined sample has the highest heat transfer coefficient compared to the oxidized samples.

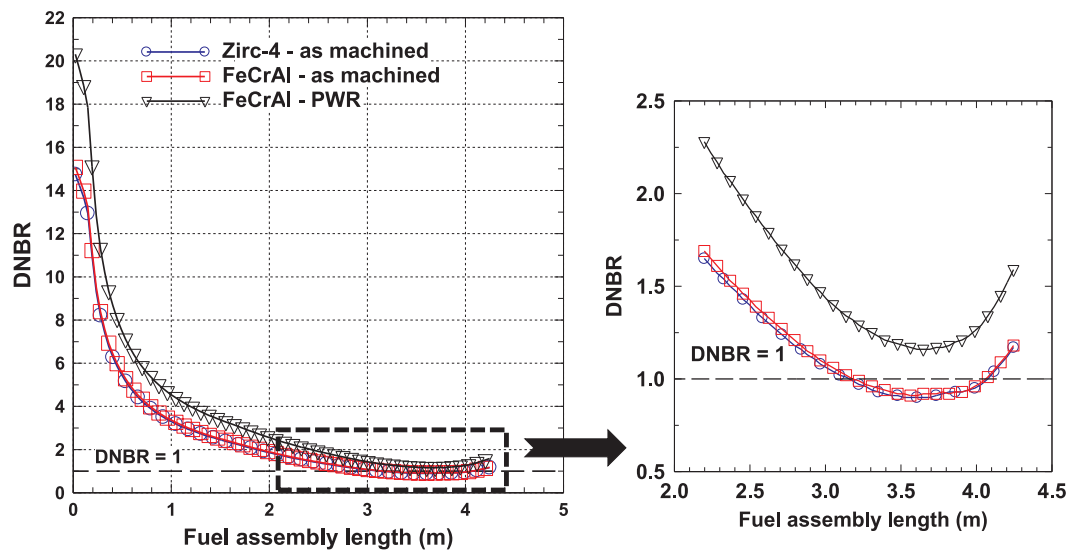


Fig. 18. DNBR axial distribution for Zirc-4 (as machined), and Fe13Cr5Al (as machined and PWR).

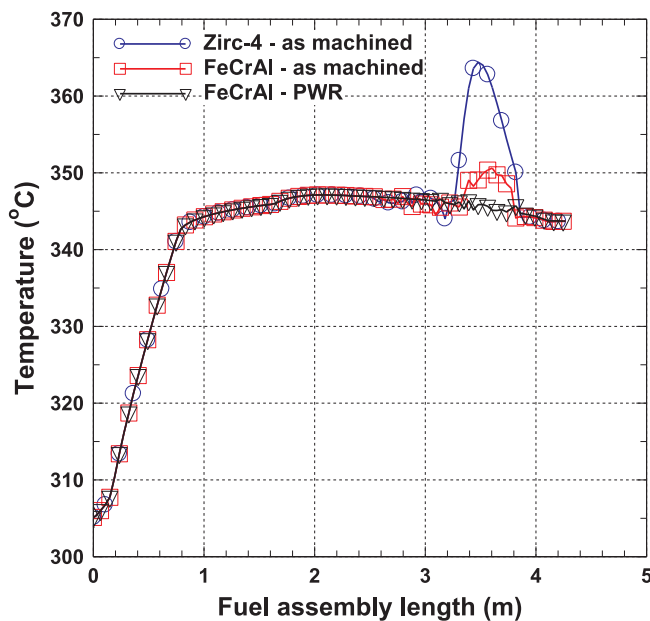


Fig. 19. Cladding outer surface temperature.

The CHF values predicted using the Kandlikar model (Eq. (5)) and the measured CHF values are compared in Fig. 13. As shown in the figure, the measured CHF for as machined samples is lower than the predicted values, but it is within the error of the model as reported by the authors (21.59%). For the oxidized samples, the predicted and measured CHF values are in agreement within $\pm 4\%$. These results show that the Kandlikar model (Eq. (5)) also can predict the CHF for as machined and oxidized Fe13Cr4Al under different simulated LWR conditions.

4.5. PWR simulation results

This section focuses on developing a rough prediction of the heat transfer characteristics of FeCrAl in a PWR core, particularly CHF and departure from nucleate boiling ratio (DNBR). The DNBR is defined as the ratio of the CHF at any location to the operating heat flux at that location. Many correlations are available in the literature to predict the CHF of zirconium alloys under flow boiling conditions, but they are not applicable to other cladding materials and must be modified. This study

focuses on the W-3 correlation developed by Tong (1967), and uses a constant multiplier to make the correlation more applicable to FeCrAl. The CHF and DNBR calculations were then performed using the Consortium for Advanced Simulation of Light Water Reactors (CASL) sub-channel code COBRA-TF (CTF) (Salko et al., 2015) for a one-eighth model of a 17×17 PWR fuel assembly.

Resulting CHF values from pool boiling experiments for as machined Zirc-4 (784.1 kW/m^2), as machined Fe13Cr5Al (807.5 kW/m^2), and Fe13Cr5Al that was oxidized under PWR conditions (1088.7 kW/m^2) were used to calculate a multiplier for each of the two FeCrAl samples, which was then implemented into the CTF source code for the W-3 CHF correlation. Each multiplier was calculated based on the delta between the measured CHF values for Zirc-4 and the Fe13Cr5Al samples. The multiplier was 1.029 for the as-machined FeCrAl and 1.388 for the oxidized FeCrAl.

4.5.1. PWR fuel assembly model

The CTF model developed for this study is an octant of a 17×17 PWR fuel assembly and is shown in Fig. 14. The model has 45 subchannels and 45 rods, 6 of which are unheated. Of the 45 rods, 17 of them (12 heated rods and 5 unheated rods) are either one-half or one-eighth rods due to being located along a line of symmetry. All inlet conditions of the model are consistent with typical PWR values except for the power, which is set much higher than typical values to approximately 4.2 MW of net power for the octant to simulate a possible accident condition. The model is based on a PWR model created by Basile et al. (1999) in COBRA-EN that also imposed the 50% overpower condition. The overpower condition was left in the current study's model to verify the CTF model against the COBRA-EN results and also to force some of the rods in the CTF model to exceed CHF.

The input for the three cladding cases was identical except for the material properties, fuel pellet diameters, and cladding thicknesses. Differences in fuel rod parameters were provided by Brown et al. (2017) and Liu et al. (2017) and are summarized in Table 3, along with some key results.

4.5.2. DNBR results

Fig. 15 presents the axial variation of CHF as predicted by the W-3 correlation for Zirc-4 as machined, as machined FeCrAl, and oxidized FeCrAl-PWR for a single fuel rod in the PWR model. CHF for Zirc-4 and as machined FeCrAl were very similar due to the multiplier being 1.029, while the oxidized FeCrAl CHF was significantly greater.

The W-3 CHF prediction is dependent on the pressure, mass flux,

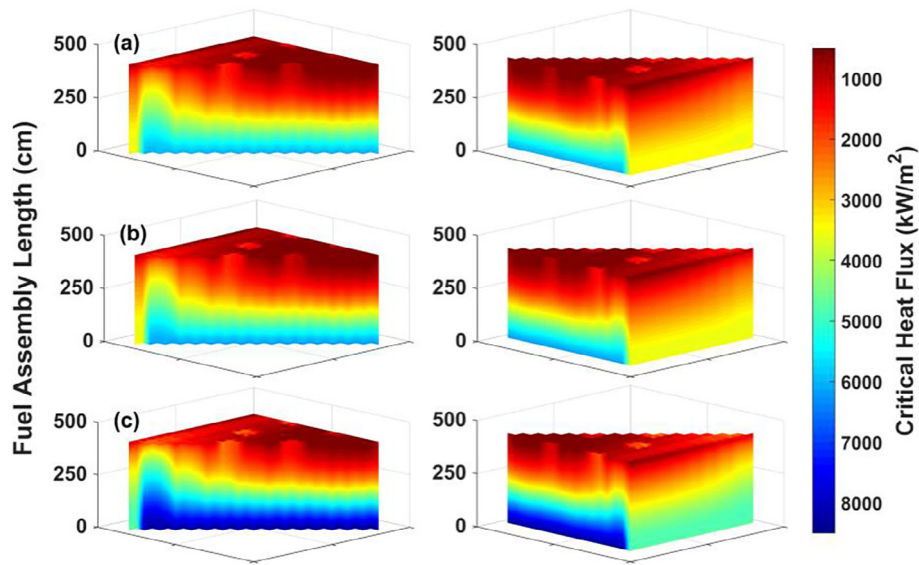


Fig. 20. Spatial Distribution of CHF for (a) Zirc-4, (b) As machined FeCrAl, and (c) FeCrAl-PWR.

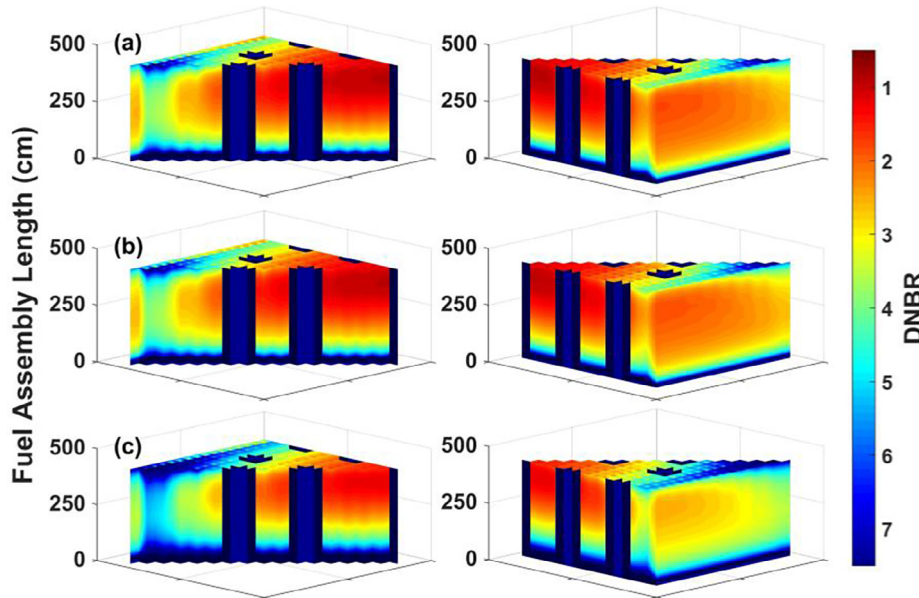


Fig. 21. Spatial Distribution of DNBR for (a) Zirc-4, (b) As machined FeCrAl, and (c) FeCrAl-PWR.

equilibrium quality, and the difference between saturation and actual enthalpy of the coolant (Tong, 1967). While the pressure and mass flux remain approximately constant throughout the length of the assembly, the equilibrium quality and enthalpy of the coolant change significantly, as shown in Figs. 16 and 17, respectively. The changes in equilibrium quality and coolant enthalpy explain the axial change in predicted CHF. The 50% overpower condition imposed on the model causes the equilibrium quality to be much higher than expected in a typical PWR.

Fig. 18 shows the axial distribution of DNBR for the same fuel rod used for predicting CHF as shown in Fig. 15. Again, the calculated DNBR for Zirc-4 and as machined FeCrAl are very similar, but the DNBR for oxidized FeCrAl was higher.

As shown in the zoomed-in Fig. 18 (right view), the overpower condition caused the DNBR for as machined Zirc-4 and FeCrAl to drop below 1.0, while the oxidized FeCrAl DNBR is always greater than 1.0. This means that the heat flux of the rod reached and exceeded CHF at distinct locations along the fuel rod, which significantly reduces the

heat transfer from the rod to the coolant. The reduction of the heat transfer can be recognized in the cladding outer surface temperature spikes as shown in Fig. 19 at the same axial location where the heat flux is equal or exceeded CHF for the Zirc-4 and as machined FeCrAl cladings.

Notice that the DNBR of the as machined Zirc-4 and FeCrAl rises back above 1.0, which is why the cladding temperatures return to the same temperature as the oxidized FeCrAl (Figs. 18 and 19). While one may not expect a rewetting of the system and a return to nucleate boiling, according to Todreas and Kazimi (2011), the W-3 correlation is only valid for equilibrium qualities in the range of -0.15 to 0.15 and an assembly length of 3.70 m. Both limits are exceeded in the PWR model, so CHF values predicted outside these limits may be incorrect.

To understand the CHF and DNBR distribution throughout the assembly, rather than on a rod-by-rod basis, a 3-D spatial distribution map was made for both parameters, as shown in Figs. 20 and 21. The CHF and DNBR distributions for the Zirc-4 and as machined FeCrAl are nearly indistinguishable from each other. The distributions for the

oxidized FeCrAl, however, show a significantly higher CHF and DNBR throughout the fuel assembly octant.

5. Discussion and conclusion

The presented results show that FeCrAl samples oxidized under simulated LWR conditions experience higher surface roughness (Fig. 3) and lower static, advance, and receding contact angles (higher wettability) (Figs. 6 and 7) compared to Zirc-4 samples oxidized under the same conditions. The oxidation rate and characteristics of the oxide layer (thickness and microstructure) could be the major factor of surface energy change and hence in the differences of surface wettability. Iron-based alloys, such as FeCrAl, have been shown to have the lower oxidation rate under high-temperature steam environments compared to Zirc-4 (Terrani et al., 2016). The reported results on the oxide layer thickness showed that Fe12Cr4Al has a thinner formed oxide layer compared to Fe13CrAl, but both have a similar oxide layer composition (Terrani et al., 2016). This difference in the oxidation layer thickness could explain the lower contact angle for Fe12Cr4Al relative to Fe13Cr5Al.

The higher average surface roughness, R_a , measured of FeCrAl samples indicates a higher wetted surface area in contact with the liquid compared to the footprint area. Such larger surface area would allow for a higher heat transfer rate of FeCrAl compared to other surfaces. In addition, the large wetted surface area is usually associated with a larger number of microcavities that could become active sites for bubble nucleation. An increased number of active bubble nucleation sites and the wetted surface area should enhance the boiling heat transfer coefficient and increase CHF. According to the Wenzel model (Wenzel, 1936), an increased surface roughness would increase the surface wettability (lower contact angle) for hydrophilic surfaces ($\theta < 90^\circ$). The high wettability promotes re-wetting of dry patches that develop on the surface in the bubble coalescence region immediately after bubble departure, which delays the occurrence of CHF. The high wettability may also allow liquid to partially and/or fully flood and deactivate the cavities on the surface and hence decrease the heat transfer coefficient.

The surface roughness measurements presented in Fig. 3 shows that the FeCrAl samples that were autoclaved and formed thin oxidation layers have a higher average surface roughness value ($0.59 \mu\text{m}$) compared to Zirc-4 ($0.36 \mu\text{m}$) alloys oxidized under the same conditions. As shown in Fig. 10, the wettability of as machined samples (no oxidation) is lower (higher contact angle) compared to the oxidized samples. Thus, the higher surface roughness and wettability (lower contact angle) explain the higher CHF and heat transfer coefficient for the as machined Fe13Cr4Al compared to the as machined Zirc-4 (Figs. 11 and 13).

Although the surface roughness was averaged for all Fe13Cr4Al samples, Fig. 3 shows that the average surface roughness for FeCrAl oxidized under PWR conditions ($R_a = 0.627 \mu\text{m}$) is slightly higher than the average surface roughness of the FeCrAl sample oxidized under BWR-NWC conditions ($R_a = 0.61 \mu\text{m}$), and both are higher than the measured roughness of the as machined FeCrAl ($R_a = 5.9 \mu\text{m}$). The wettability of PWR-oxidized samples is much higher than for the BWR-NWC-oxidized samples, and the as machined samples have the lowest wettability (higher contact angle). The slight difference in the average surface roughness values and wettability between these samples would explain the higher measured CHF for PWR and BWR-NWC-oxidized samples compared to the as machined samples and vice versa for the heat transfer coefficient. The measured pool boiling CHF of selected oxidized Fe13Cr5Al and as machined samples are in a good agreement with the predicted CHF values calculated using Kandlikar (2001) model (Eq. (5)) as shown in Fig. 13.

A rough estimate of CHF and DNBR for FeCrAl claddings in a PWR under a 50% overpower condition was calculated using the experimentally measured pool boiling CHF results to modify the W-3 correlation. Although pool boiling CHF does not directly translate to flow

boiling CHF, these results provide a first estimate of the thermal hydraulic performance of FeCrAl cladding under an overpower condition.

The as machined Zirc-4 and as machined FeCrAl fuel rods reach a DNBR less than or equal to 1.0. This means that the heat flux through the cladding wall has exceeded the Critical Heat Flux, and a thin vapor film has developed between the cladding wall and the liquid water. On the other hand, the oxidized FeCrAl DNBR was always greater than 1.0, meaning CHF was never reached and overheating of the fuel and cladding is not an issue. Although Zirc-4 and FeCrAl melting temperatures were not reached in this case, Fig. 19 shows how the reduction in heat transfer quickly results in a temperature increase in the cladding. The DNBR for as machined Zirc-4 and FeCrAl later rises above 1.0 and the cladding temperature returns to normal, but this may be caused by a breakdown in the W-3 correlation due to exceeding equilibrium quality and flow length limits. These preliminary results indicate that oxidized FeCrAl may be able to withstand these accident conditions without leading to a boiling crisis.

6. Acknowledgments

This research was funded by Oak Ridge National Lab (USA), Idaho National Lab (USA), and Nuclear Energy University Program (NEUP) award (#17-12688) (USA).

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